Safety data sheet

1. Substance/preparation and manufacturer/supplier identification

Tinosan® HP 100

Use: Biocide active ingredient

Manufacturer/supplier:
BASF SE
67056 Ludwigshafen
GERMANY
Operating Division Care Chemicals
Telephone: +49 621 60-57579
E-mail address: emd-ems-ehs-masterdata@basf.com

Emergency information:
International emergency number:
Telephone: +49 180 2273-112

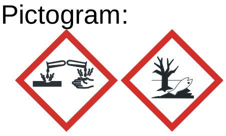
2. Hazard identification

Classification according to UN GHS 2009

Classification of the substance and mixture:
Serious eye damage/eye irritation: Cat. 1
Hazardous to the aquatic environment - acute: Cat. 1
Hazardous to the aquatic environment - chronic: Cat. 1

Label elements and precautionary statement:

Pictogram:



Signal Word:
Danger

Hazard Statement:

H318	Causes serious eye damage.
H400	Very toxic to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.

Precautionary Statements (Prevention):

P273	Avoid release to the environment.
P280	Wear eye/face protection.

Precautionary Statements (Response):

P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P310	Immediately call a POISON CENTER or doctor/physician.
P391	Collect spillage.

Precautionary Statements (Disposal):

P501	Dispose of contents/container to hazardous or special waste collection point.

Other hazards which do not result in classification:
No specific dangers known, if the regulations/notes for storage and handling are considered.

3. Composition/information on ingredients

Chemical nature

Preparation based on: propane-1,2-diol, Phenol, 5-chloro-2-(4-chlorophenoxy)-

Hazardous ingredients

Phenol, 5-chloro-2-(4-chlorophenoxy)-

Content (W/W): >= 25 % - < 50 % CAS Number: 3380-30-1	Eye Dam./Irrit.: Cat. 1 Aquatic Acute: Cat. 1 Aquatic Chronic: Cat. 1 M-factor acute: 10 M-factor chronic: 10

Phenol, 5-chloro-2-(2,4-dichlorophenoxy)-

Content (W/W): >= 0 % - < 0.1 % CAS Number: 3380-34-5	Skin Corr./Irrit.: Cat. 2 Eye Dam./Irrit.: Cat. 2A Aquatic Acute: Cat. 1 Aquatic Chronic: Cat. 1 M-factor acute: 100 M-factor chronic: 1

4. First-Aid Measures

General advice:
Remove contaminated clothing.

If inhaled:
Immediately administer a corticosteroid from a controlled/metered dose inhaler.

On skin contact:
Immediately wash thoroughly with plenty of water, apply sterile dressings, consult a skin specialist.

On contact with eyes:
Immediately wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Note to physician:
Symptoms: Additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far
Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Suitable extinguishing media:
water spray, dry powder, foam

Unsuitable extinguishing media for safety reasons:
water jet

Specific hazards:
harmful vapours
Evolution of fumes/fog. The substances/groups of substances mentioned can be released in case of fire.

Special protective equipment:
Wear a self-contained breathing apparatus.

Further information:
The degree of risk is governed by the burning substance and the fire conditions. Contaminated extinguishing water must be disposed of in accordance with official regulations.

6. Accidental Release Measures

Personal precautions:
Use personal protective clothing. Information regarding personal protective measures, see section 8. Do not breathe vapour/spray.

Environmental precautions:
Contain contaminated water/firefighting water. Do not discharge into drains/surface waters/groundwater.

Methods for cleaning up or taking up:
For large amounts: Dike spillage. Pump off product.
For residues: Pick up with suitable absorbent material.
Dispose of absorbed material in accordance with regulations.

7. Handling and Storage

Handling

Avoid vapor formation. Keep away from sources of ignition - No smoking. Handle in accordance with good industrial hygiene and safety practice.

Protection against fire and explosion:
Take precautionary measures against static discharges.

Storage

Suitable materials for containers: High density polyethylene (HDPE)
Further information on storage conditions: Keep container tightly closed and dry; store in a cool place.

Storage stability:
Storage temperature: -10 - 40 °C

Protect from temperatures below: -10 °C
The packed product is destroyed at low temperatures or by frost.
Protect from temperatures above: 40 °C

8. Exposure controls and personal protection

Components with occupational exposure limits

No occupational exposure limits known.

Personal protective equipment

Respiratory protection:
Suitable respiratory protection for higher concentrations or long-term effect: Gas filter for gases/vapours of organic compounds (boiling point >65 °C, e. g. EN 14387 Type A)

Hand protection:
Chemical resistant protective gloves
Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN 374):
nitrile rubber (NBR) - 0.4 mm coating thickness

Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing. Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:
Tightly fitting safety goggles (cage goggles) (e.g. EN 166) and face shield.

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures:
Wearing of closed work clothing is recommended. No eating, drinking, smoking or tobacco use at the place of work. Handle in accordance with good industrial hygiene and safety practice.

9. Physical and Chemical Properties

Form:	liquid	
Colour:	colourless to brownish	
Odour:	product specific	
Odour threshold:	Not determined due to potential health hazard by inhalation.	
pH value:	6.5	
pour point:	-40 °C	(DIN ISO 3016)
Boiling point:	approx. 184 °C	(estimated)
Information on: propane-1,2-diol		
Boiling point:	184 °C (1,003.2 hPa)	(Directive 92/69/EEC, A.2)

Flash point:	99 °C	(DIN EN 22719; ISO 2719)
Evaporation rate:	Value can be approximated from Henry's Law Constant or vapor pressure.	
Flammability (solid/gas):	not determined	
Lower explosion limit:	For liquids not relevant for classification and labelling., The lower explosion point may be 5 - 15 °C below the flash point.	
Upper explosion limit:	For liquids not relevant for classification and labelling.	
Ignition temperature:	410 °C	
Thermal decomposition:	330 °C	(DSC (DIN 51007))

Self ignition: not self-igniting

Self heating ability: not applicable, the product is a liquid

Explosion hazard: not explosive
Fire promoting properties: not fire-propagating

Vapour pressure: 0.22 hPa (OECD Guideline 104)

Density: 1.13 g/cm3 (25 °C) (DIN 51757)
1.11 g/cm3 (35 °C) (DIN 51757)

Relative density:
No data available.

Bulk density:
not applicable

Relative vapour density (air):
not determined

Solubility in water: insoluble
Hygroscopy: not applicable
Solubility (qualitative) solvent(s): Alkalines, Ethanol, isopropanol
soluble
Solubility (qualitative) solvent(s): aromatic hydrocarbons, non-polar solvents
partly soluble
Solubility (qualitative) solvent(s): inorganic acids, organic solvents
insoluble
Solubility (qualitative) solvent(s): mineral oils
soluble, cloudy
Partitioning coefficient n-octanol/water (log Pow):
Study does not need to be conducted.

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-
Adsorption/water - soil: KOC: 1412.54; log KOC: 3.15 (OECD Guideline 121)

Surface tension:
No data available.

Viscosity, dynamic: 80 mPa.s (25 °C) (DIN 54453)
60 mPa.s (30 °C) (DIN EN 12092)

Other Information:
If necessary, information on other physical and chemical parameters is indicated in this section.

10. Stability and Reactivity

Conditions to avoid:
See SDS section 7 - Handling and storage.

Thermal decomposition: 330 °C (DSC (DIN 51007))

Substances to avoid:
chlorinating agents

Hazardous reactions:
No hazardous reactions when stored and handled according to instructions.

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological Information

Acute toxicity

Experimental/calculated data:
ATE rat (oral): > 5,000 mg/kg (other)

ATE (by inhalation): > 20 mg/l
Determined for vapor

ATE (by inhalation): > 5 mg/l
Determined for mist

ATE (dermal): > 5,000 mg/kg

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-
Assessment of acute toxicity:
Virtually nontoxic after a single skin contact. Virtually nontoxic after a single ingestion.

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-
Experimental/calculated data:
LD50 rat (oral): > 2,000 mg/kg (OECD Guideline 423)

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-
Experimental/calculated data:
LD50 rat (dermal): > 2,000 mg/kg (OECD Guideline 402)

Irritation

Assessment of irritating effects:
May cause severe damage to the eyes.

Experimental/calculated data:
Skin corrosion/irritation rabbit: non-irritant (OECD Guideline 404)

Serious eye damage/irritation rabbit: irreversible damage (OECD Guideline 405)

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-
Assessment of irritating effects:
Not irritating to the skin. May cause severe damage to the eyes.

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-
Experimental/calculated data:
Skin corrosion/irritation rabbit: non-irritant (OECD Guideline 404)

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-
Experimental/calculated data:
Serious eye damage/irritation rabbit: irreversible damage (OECD Guideline 405)

Respiratory/Skin sensitization

Experimental/calculated data:
guinea pig: Non-sensitizing. (OECD Guideline 406)

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-
Assessment of sensitization:
Skin sensitizing effects were not observed in animal studies.

Germ cell mutagenicity

Assessment of mutagenicity:
The substance was not mutagenic in bacteria. The substance was mutagenic in a mammalian cell culture test system. The substance was not mutagenic in studies with mammals.

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-
Assessment of mutagenicity:
The substance was not mutagenic in bacteria. A mutagenicity test using mammalian cells yielded equivocal results. However, structurally related substances were not mutagenic. The substance was not mutagenic in a test with mammals.

Carcinogenicity

Assessment of carcinogenicity:
Based on available Data, the classification criteria are not met.

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-
Assessment of carcinogenicity:
In long term studies in mice in which the substance was given by feed, a carcinogenic effect was observed. The effect is caused by an animal specific mechanism that has no human counter part. The whole of the information assessable provides no indication of a carcinogenic effect. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Reproductive toxicity

Assessment of reproduction toxicity:
Based on available Data, the classification criteria are not met.

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-
Assessment of reproduction toxicity:
The results of animal studies gave no indication of a fertility impairing effect. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Developmental toxicity

Assessment of teratogenicity:
Based on available Data, the classification criteria are not met.

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-
Assessment of teratogenicity:
In animal studies the substance did not cause malformations. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Specific target organ toxicity (single exposure):

Assessment of STOT single:
Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
Based on available Data, the classification criteria are not met.

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-
Assessment of repeated dose toxicity:
The substance may cause damage to the liver after repeated ingestion of high doses, as shown in animal studies. The substance may cause damage to the kidney after repeated ingestion.

Aspiration hazard

No aspiration hazard expected.

Other relevant toxicity information

The product has not been tested. The statements on toxicology have been derived from the properties of the individual components.

12. Ecological Information

Ecotoxicity

Assessment of aquatic toxicity:

Very toxic (acute effect) to aquatic organisms. May cause long-term adverse effects in the aquatic environment. Depending on local conditions and existing concentrations, disturbances in the biodegradation process of activated sludge are possible.

Toxicity to fish:
LC50 (96 h) <= 1 mg/l, Fish (OECD 203; ISO 7346; 92/69/EEC, C.1, static)
The statement of the toxic effect relates to the analytically determined concentration. Toxic effects occur within the range of solubility.

Aquatic invertebrates:
EC50 (48 h) <= 1 mg/l, Daphnia magna (OECD Guideline 202, part 1, static)
The details of the toxic effect relate to the nominal concentration. Toxic effects occur within the range of solubility.

Aquatic plants:
EC50 (72 h) <= 1 mg/l (growth rate), Desmodesmus subspicatus (OECD Guideline 201, static)
The details of the toxic effect relate to the nominal concentration. Toxic effects occur within the range of solubility.

No observed effect concentration (72 h) <= 0.1 mg/l (growth rate), Desmodesmus subspicatus (OECD Guideline 201, static)
The statement of the toxic effect relates to the analytically determined concentration.

Microorganisms/Effect on activated sludge:
EC50 (3 h) approx. 25 mg/l, activated sludge (OECD Guideline 209)

Chronic toxicity to fish:
No observed effect concentration <= 0.1 mg/l, Fish
The statement of the toxic effect relates to the analytically determined concentration.

Chronic toxicity to aquatic invertebrates:
The ecological data given are those of the active ingredient.

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-
Toxicity to fish:
LC50 (96 h) 0.7 mg/l, Brachydanio rerio (OECD 203; ISO 7346; 92/69/EEC, C.1, static)
The statement of the toxic effect relates to the analytically determined concentration. Toxic effects occur within the range of solubility.

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-
Aquatic invertebrates:
EC50 (48 h) 0.32 mg/l, Daphnia magna (OECD Guideline 202, part 1, static)
The details of the toxic effect relate to the nominal concentration. Toxic effects occur within the range of solubility.

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-
Aquatic plants:
EC50 (72 h) 0.038 mg/l (growth rate), Desmodesmus subspicatus (OECD Guideline 201, static)
The details of the toxic effect relate to the nominal concentration. Toxic effects occur within the range of solubility.

No observed effect concentration (72 h) 0.0093 mg/l (growth rate), Desmodesmus subspicatus (OECD Guideline 201, static)
The statement of the toxic effect relates to the analytically determined concentration.

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-
Microorganisms/Effect on activated sludge:
EC50 (3 h) 8 mg/l, activated sludge, domestic (OECD Guideline 209, static)
The details of the toxic effect relate to the nominal concentration.

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-
Chronic toxicity to fish:
No observed effect concentration (96 d) 34,1 µg/l, Oncorhynchus mykiss (OPP 72-4 (EPA-Guideline), Flow through.)
The statement of the toxic effect relates to the analytically determined concentration. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-
Chronic toxicity to aquatic invertebrates:
No observed effect concentration (21 d), 0.22 mg/l, Daphnia magna (OECD Guideline 211, semistatic)

Assessment of terrestrial toxicity:
No data available concerning terrestrial toxicity.

Soil living organisms:
The ecological data given are those of the active ingredient.

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-
Soil living organisms:
LC50 (14 d) 693 mg/kg, Eisenia foetida (OECD Guideline 207, artificial soil)
The details of the toxic effect relate to the nominal concentration.

Mobility

Assessment transport between environmental compartments:
The ecological data given are those of the active ingredient.

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-
Assessment transport between environmental compartments:
The substance will not evaporate into the atmosphere from the water surface.
Adsorption to solid soil phase is expected.

Persistence and degradability

Assessment biodegradation and elimination (H2O):
The ecological data given are those of the active ingredient.

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-
Elimination information:
50 - 60 % CO_2 formation relative to the theoretical value (61 d) (OECD 301B; ISO 9439; 92/69/EEC, C.4-C) (aerobic, activated sludge, domestic, non-adapted)

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-
Assessment of stability in water:
In contact with water the substance will hydrolyse slowly.

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-
Information on Stability in Water (Hydrolysis):
$t_{1/2}$ > 1 a (pH 7, 25 °C, pH value 7), (OECD Guideline 111)

Bioaccumulation potential

Assessment bioaccumulation potential:
The ecological data given are those of the active ingredient.

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-
Assessment bioaccumulation potential:
Does not significantly accumulate in organisms.

Information on: Phenol, 5-chloro-2-(4-chlorophenoxy)-
Bioaccumulation potential:
Bioconcentration factor: 68 - 77 (28 d), Cyprinus carpio (OECD Guideline 305 E)

Additional information

Add. remarks environm. fate & pathway:
Treatment in biological waste water treatment plants has to be performed according to local and administrative regulations.

Other ecotoxicological advice:
The product has not been tested. The statements on ecotoxicology have been derived from the properties of the individual components. Do not allow to enter soil, waterways or waste water channels.

13. Disposal Considerations

Must be disposed of or incinerated in accordance with local regulations.

Contaminated packaging:
Uncontaminated packaging can be re-used.
Packs that cannot be cleaned should be disposed of in the same manner as the contents.

14. Transport Information

Domestic transport:

Packing group:	III
ID number:	UN 3082
Transport hazard class(es):	9, EHSM
Proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (contains DIPHENYL ETHER)

Sea transport
IMDG

Packing group:	III
ID number:	UN 3082
Transport hazard class(es):	9, EHSM
Marine pollutant:	YES
Proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (contains DIPHENYL ETHER)

Air transport
IATA/ICAO

Packing group:	III
ID number:	UN 3082
Transport hazard class(es):	9, EHSM
Proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (contains DIPHENYL ETHER)

15. Regulatory Information

<u>Other regulations</u>

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

16. Other Information

This product is of industrial quality and unless otherwise specified or agreed intended exclusively for industrial use. This includes the mentioned and recommended usage. Any other intended applications should be discussed with the manufacturer. In particular this concerns the application for products that are the object of special standards and regulations.

Vertical lines in the left hand margin indicate an amendment from the previous version.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.